[Re] A Boolean network inference from time-series gene expression data using a genetic algorithm

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Abstract

This article reports a clean-room reimplementation of GABNI, a two-stage Boolean network inference method combining mutual information-based Boolean network inference (MIBNI) with a genetic algorithm (GA). The accompanying Python package implements data loading, optional binarization, MIBNI candidate selection, GA-based regulator search, Boolean transition prediction, edge sign assignment, and structural/dynamical evaluation. On the yeast cell-cycle benchmark distributed with the repository, the reimplementation reproduces perfect one-step dynamics accuracy and obtains a sparse inferred network with TP=15, FP=2, and FN=14 under the directed unsigned structural protocol. These structural counts differ from the original paper's reported yeast result, which we attribute mainly to short time-series under-determination, parsimony pressure in the reimplementation, and missing low-level details of the original implementation. The submission therefore supports the main dynamical claim while documenting the limits of exact structural replication. We provide a clean-room reimplementation of GABNI and reproduce selected results from Barman and Kwon (2018).

1 Introduction

Barman and Kwon introduced GABNI, a Boolean network inference method for time-series gene expression data. The method combines a fast MIBNI stage with a GA stage that searches regulator sets and Boolean update rules when the first stage does not find an optimal solution. The original article reports strong structural and dynamical accuracy on artificial and real gene-expression benchmarks, including a yeast cell-cycle case study.

The purpose of this replication is narrower than a full re-run of every experiment in the original article. The accompanying repository provides a clean-room Python reimplementation of the core GABNI pipeline and a reproducible yeast case study. This article summarizes what is reproduced, where the results diverge, and which assumptions are likely responsible for the divergence.

2 Original method

The original GABNI workflow can be summarized as follows:

1. convert expression observations into Boolean states where necessary;
2. for each target gene, use mutual information to restrict candidate regulators;
3. apply MIBNI to search for a Boolean update rule in smaller candidate spaces;
4. if MIBNI fails to achieve an optimal trajectory match, apply a GA over regulator selections and literal polarities;
5. combine per-gene regulatory rules into a final Boolean network;
6. evaluate the inferred network using structural accuracy and one-step dynamics consistency.

3 Reimplementation

The reproduction repository is available at https://github.com/Bonartze/gabni-reproduction. It exposes a command-line interface named `gabni` and includes the yeast cell-cycle time-series data and a signed directed gold network. The implementation is intentionally self-contained and does not depend on source code from the original authors.

The main implementation choices are:

- Python package with dependencies limited to NumPy, pandas, and scikit-learn;
- optional KMeans binarization for continuous input data;
- MIBNI candidate filtering based on lag-1 mutual information;
- GA chromosomes encoding positive and negated regulator literals;
- deterministic seeds for reproducible runs;
- structural metrics in both directed unsigned and directed signed forms;
- optional pruning and exact minimization utilities to explore sparse regulator sets.

4 Reproducibility protocol

The yeast reference run can be reproduced from a clean environment with:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e .
mkdir -p results
gabni --seed 0 --force-ga --no-exact-minimize --prune \
--max-reg-frac 0.4 \
--input datasets/yeast_cell_cycle_timeseries_s2.csv \
--gold_edges datasets/yeast_cell_cycle_gold_edges_no_self.csv \
--out_edges results/pred_edges_seed0.csv \
--out_summary results/summary_seed0.json
```

The command writes the inferred edge list and a JSON summary. The summary includes the number of genes and time points, mean gene consistency, one-step dynamics accuracy, number of predicted edges, and structural comparison against the gold network.

5 Results

Table 1 compares the reported yeast case study numbers with the reference run from this reimplementation. The reimplementation achieves perfect dynamics accuracy, matching the most important dynamical result. The inferred structure is sparser than the network reported in the original article: it has fewer false positives, but also fewer true positives and therefore more false negatives.

Source	Dynamics accuracy	TP	FP	FN
Original paper, yeast, 0% noise	1.0	18	3	11
This reimplementation, reference run	1.0	15	2	14

Table 1: Yeast cell-cycle benchmark comparison under the directed unsigned structural protocol.

6 Discrepancies and limitations

Exact structural reproduction is difficult for this benchmark because the yeast time series contains only 13 time points for 11 genes. Many regulator sets can explain the same observed transitions, so perfect one-step dynamics does not uniquely determine the network topology. The reimplementation also applies a parsimony-oriented search configuration in the reference run, including a compact regulator set preference and optional pruning. This makes the inferred network more conservative and explains the lower false-positive count together with lower recall.

A second limitation is that the original paper does not fully specify every low-level implementation detail needed for bit-exact reproduction. For example, GA selection, tie-breaking, repair, stopping behavior, and post-processing choices can alter the final structural edge set while preserving trajectory accuracy. The submitted code therefore aims to reproduce the algorithmic idea and the dynamical behavior, rather than claiming exact recovery of the original implementation.

7 Provenance and acknowledgements

This work was motivated by ReScience C call-for-replication issue #7, where `@Caiofcas` proposed the replication target and noted that no original implementation was available. I thank `@Caiofcas` for identifying the replication target and the ReScience C editors for maintaining the call-for-replication process.

8 Conclusion

The reproduction supports the core dynamical result of GABNI on the yeast cell-cycle benchmark: the inferred Boolean network can reproduce the observed one-step dynamics with accuracy 1.0. The structural result is close but not identical to the original report, with a sparser inferred network. The discrepancy is documented and reproducible, and the submitted repository provides commands and outputs for independent verification.

References

[1] Barman, S. and Kwon, Y.-K. (2018). A Boolean network inference from time-series gene expression data using a genetic algorithm. *Bioinformatics*, 34(17), i927–i933. doi:10.1093/bioinformatics/bty584.

[2] Barman, S. and Kwon, Y.-K. (2017). A novel mutual information-based Boolean network inference method from time-series gene expression data. *PLOS ONE*, 12(2), e0171097. doi:10.1371/journal.pone.0171097.